

Space-borne frequency comb metrology

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Precision time references in space are of major importance to satellite-based fundamental science, global satellite navigation, earth observation, and satellite formation flying. Here we report on the operation of a compact, rugged, and automated optical frequency comb setup on a sounding rocket in space under microgravity. The experiment compared two clocks, one based on the optical D_2 transition in Rb, and another on hyperfine splitting in Cs. This represents the first frequency comb based optical clock operation in space, which is an important milestone for future satellite-based precision metrology. Based on the approach demonstrated here, future space-based precision metrology can be improved by orders of magnitude when referencing to state-of-the-art optical clock transitions. © 2016 Optical Society of America

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1. INTRODUCTION

The upcoming generation of optical atomic frequency references has shown an enormous increase in accuracy and stability compared to the present official atomic clock standard based on Cesium (see, e.g., [1,2]). These amazingly stable optical atomic clocks are based on energy transitions that have been carefully quantum engineered and controlled, and that are oscillating at frequencies of several hundred terahertz with a stability of 10^{-18} . Optical clocks have, for example, demonstrated a precision that is sensitive enough to detect a few centimeters of elevation change within the Earth's gravitational potential, caused by the relativistic redshift. In addition, the recent discovery of gravitational waves [3] has triggered a growing interest in their advanced detection. Gravitational wave detectors are large Michelson interferometers that compare optical phases of a stable laser sent via the two orthogonal arms. Typical laser linewidths are ≈ 10 Hz, and a strain sensitivity approaching 10^{-23} Hz $^{-1/2}$ has been reported [4]. Advanced gravitational wave detectors like LISA, which are designed as very large baseline interferometers between satellites, have been a long-lasting dream of the scientific community [5]. Such a space-based device would be important for the detection of low-frequency waves

(<1 Hz). A large-scale gravitational wave detector satellite mission called eLISA/NGO (see e.g., [6]) is currently being discussed.

Whenever it comes to the calibration and stabilization of lasers used for such precision metrology, it is often necessary to divide their optical frequency with an optical clockwork into a countable radio frequency, so that they can be referenced to an SI frequency standard. The established method of choice for this frequency division is using an optical frequency comb. Since its invention [7], the frequency comb has enabled various fundamental experiments and paved the way to groundbreaking applications [8], ranging from precision astronomy [9], femtosecond precision time transfer [10], nanometer ranging [11], fundamental physics [12–15], and gravimetry [16] to laser stabilization and cooling [17]. Apart from the work in [18], however, the operation of combs has been limited to research laboratories providing well-controlled environmental conditions. For many reasons there exists fundamental interest in applying precision laser metrology and optical clocks in space. In this environment, an optically referenced comb with ultimate performance in terms of phase noise and stability could be used to investigate general relativity [19], provide low-noise microwaves for synthetic aperture radar, calibrate lidar lasers for an in-depth study of the planetary atmosphere [20], improve

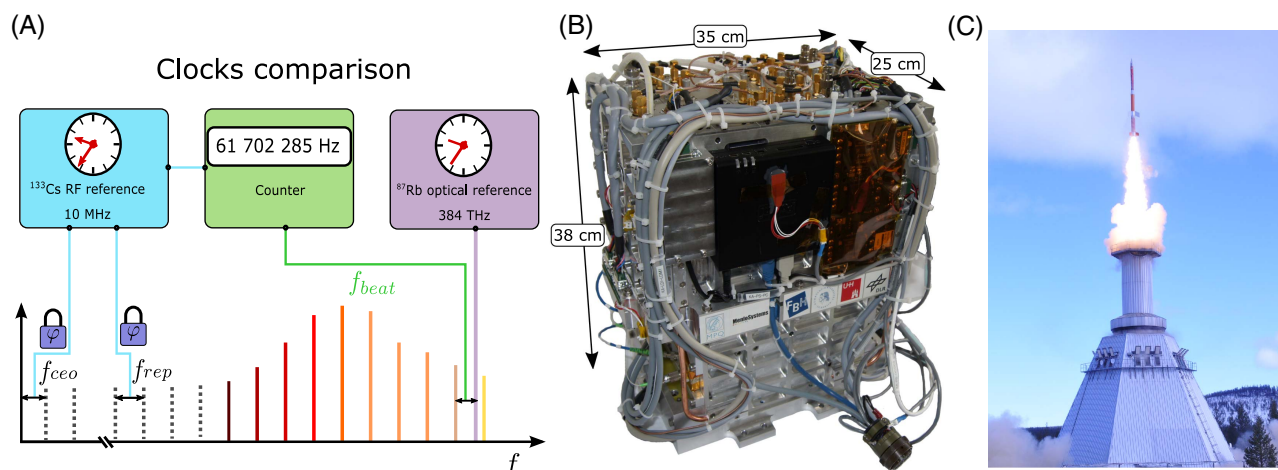


Fig. 1. (A) Schematic illustration of the clock comparison experiment performed for 360 s of microgravity during a sounding rocket flight. The frequency comb is phase stabilized to a ^{133}Cs -based atomic chip-scale clock providing 10 MHz radio frequency. A laser frequency is stabilized to an optical transition in ^{87}Rb and compared to one comb line. (B) Picture of the experimental module including the frequency comb, the Rb spectroscopy, electronics, and a control computer, (C) picture of the launch of TEXUS 51 exiting the launch tower at ESRANGE, Sweden (23 April 2015, picture provided by Airbus D&S).

on global satellite navigation systems [21], enable fast precision inter-satellite laser ranging systems for swarm navigation [11], and transfer time and frequency over large grids with femtosecond jitter [10].

Here, we demonstrate that optical frequency combs referencing a selected optical atomic line to a radio frequency clock can be scaled down to a module with a volume below 20 l, having a mass of 22 kg, and operating under the conditions of a sounding rocket flight. A schematic of our experiment, where we compare the ^{87}Rb D_2 transition at 384 THz to a 10 MHz Cs-based atomic reference, is shown in Fig. 1(A). The assembled module is depicted in Fig. 1(B). Taking into account the actual flight data and currently ongoing developments, future satellite applications become well within reach. Low-bandwidth remote operation with a high degree of automation and environmental stability against accelerations >10 g or shocks are demonstrated here. Precision thermal stabilization of the comb module with stabilities within ± 30 mK in a convection-limited microgravity environment has been achieved.

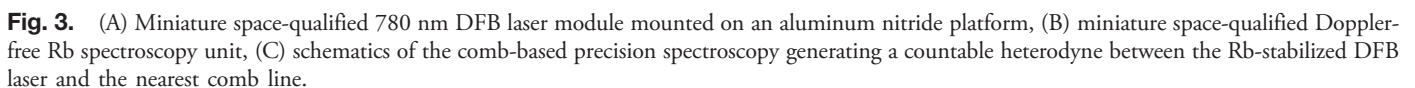
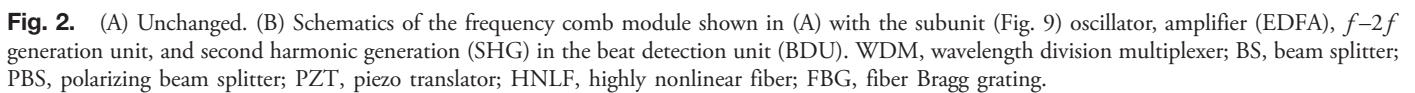
2. EXPERIMENT

Essentially, optical frequency combs are precision-controlled femtosecond lasers [7,8]. Phase-locked loops lock two radio frequencies, the pulse repetition rate f_{rep} and the carrier envelope offset frequency f_{ceo} of them. An optical spectrum is then generated with evenly spaced lines, following the comb equation $f_c = f_{\text{ceo}} + n \cdot f_{\text{rep}}$ (with n being a natural number). Usually, such lasers are mechanically sensitive to vibrations and temperature variations. This environmental sensitivity can be reduced using polarization-maintaining fibers [18,22]. The laser robustness achieved in our work is further discussed in Section 2 of Supplement 1. Our frequency comb laser is based on a nonlinear amplified loop mirror, with erbium as the gain medium [23,24]. A recent comparative study has pointed out the superior performance of this technology [25]. The complete $f_{\text{rep}} = 100$ MHz comb fits into a small package of $220 \text{ mm} \times 142 \text{ mm} \times 25 \text{ mm}$ (see Fig. 2), including the laser oscillator, erbium-doped fiber amplifier (EDFA), ridge

waveguide WG) periodically poled lithium niobate (PPLN) based $f-2f$ interferometer, and, for Rb optical beat generation at 780 nm, a 1560 nm PPLN second-harmonic generation (SHG) unit. The module has a mass of 1.65 kg, mostly due to the aluminum body, which also serves as a thermal mass. Efficient heat transport under microgravity was ensured by embedding the fiber components in space-grade silicon rubber (NUSIL CV-2500).

The comb receives 780 nm light from the atomic spectroscopy unit consisting of a micro-integrated and space-qualified diode laser locked to a Doppler-free Rb spectroscopy cell. The size of this module is $220 \text{ mm} \times 142 \text{ mm} \times 50 \text{ mm}$, with a total weight of 2.1 kg. The diode laser is a ridge waveguide distributed feedback (DFB) diode laser with a cavity length of 1.5 mm, as described in [26] and shown in Fig. 3(A). It is mounted on aluminum nitride, omitting any movable parts, including an optical micro-isolator with 60 dB isolation and a fiber coupler [27] connecting the laser to the miniature Rb spectroscopy unit, shown in Fig. 3(B). The DFB laser emits 10 mW at 780 nm behind the micro-isolator [27] and is continuously tunable over 0.7 nm via injection current control with a side mode suppression ratio of >50 dB at 10 pm resolution bandwidth. A short term laser linewidth (10 μs) of 1 MHz FWHM and a white noise floor based Lorentzian linewidth of 66 kHz has been observed. The Doppler-free spectroscopy for frequency stabilization of the DFB laser has been realized in a miniaturized optics unit situated on a Zerodur bench [28]. The Rb transition selected for the observation is the D_2 transition from the $5^2S_{1/2}$ ground state to the $5^2P_{3/2}$ excited state. Using Doppler-free saturation spectroscopy absorption lines on the order of the natural linewidth of Rb (6 MHz) can be addressed [29,30]. Frequency modulation of the DFB laser is used for locking [31,32], shown schematically in Fig. 3(C).

An automated procedure has been developed to achieve autonomous locking to a specific line during flight. The laser is first tuned in to the center between the Rb transition groups and locked to the smoothly varying error signal background in this region. There, the laser is localized within a 50 MHz window. Then, a predefined current jump is applied. This suffices to repeatedly tune the laser to the line of choice and lock to it.



an airtight vessel and pressurized to 1200 mbars to preserve constant pressure conditions during the space flight. During flight the pressure inside the vessel changed slowly by 1.6% due to an overall temperature increase of 2°C caused by heat dissipation of the device. The thermal load of the module is dumped into the baseplate of the vessel with five heatpipes and direct mechanical contact. Until 10 s before launch, the baseplate is water cooled to 15°C with a thermo-electrical chiller situated within the launch tower and connected to the payload with two removable hoses. Before launch the module was environmentally tested on a shaker table and in a climate chamber, which ensured full compliance with the expected rocket launch and reentry conditions (see Section 2 of [Supplement 1](#)). Furthermore, in an earlier work

[34] we investigated fiber degradation due to cosmic radiation, which we can exclude for such short flight durations, as is discussed in more detail in Section 3 of Supplement 1. It should be mentioned in this context that an Er-based femtosecond fiber laser has been operated recently for a duration exceeding 1 year in outer space [35]. While radiation-induced fiber darkening caused an 8.6% reduction in laser output power after 1 year, stable mode locking was achieved over the complete mission.

3. RESULTS

Two successful flights on sounding rockets have been performed up to now, TEXUS 51 in April 2015 and TEXUS 53 in January 2016, both launched from the ESRANGE Space Center [TEXUS 51 launch shown in Fig. 1(C)]. Table 1 shows some of the flight parameters for both flights. In the following discussion we concentrate primarily on the flight data of TEXUS 51, because the clock comparison data for TEXUS 53 was affected by a thermally dependent etalon in the Rb spectroscopy module, which would require advanced numerical treatment of the data to be further considered. About 5 h before launch the unit was powered and allowed to thermalize. Comb and Rb spectroscopy were locked and ground station data were acquired for reference. At $g = 9.81 \text{ m/s}^2$ the selected Rb D₂ line was determined to be

Table 1. Sounding Rocket Flight Parameters

Description	TEXUS 51	TEXUS 53
Launch date	23/04/2015	23/01/2016
Launch time (LT)	9:35 a.m.	9:30 a.m.
First stage peak thrust	8.1 g	8.3 g
Second stage peak thrust	12.6 g	12.1 g
Burnout first stage	12.1 s	11.9 s
Burnout second stage	43.2 s	43.4 s
Spin at burnout	2.8 Hz	2.8 Hz
Roll after yoyo despin	26.2°/s	18.7°/s
Micro-g phase start	73 s	73 s
Micro-g phase stop	442 s	440 s
Reentry decelerations	<13.9 g	<15.4 g
Sink rate	7.6 m/s	7.8 m/s
Impact decelerations	<33.6 g	<8.5 g
Slant range distance	76.2 km	67.9 km
Apogee	258 km	252.6 km
Payload recovery time	1:50 h	5:30 h

384.2279817(4) THz. A laser spectroscopy scan at ground is shown in Fig. 4(A). 10 s before rocket launch the system was switched into a secure state: while the Rb spectroscopy remained locked to the atomic line, the comb was unlocked in repetition rate and offset frequency to relax the servo actuators. The servo bandwidth and actuator strength are insufficient to cope with the accelerations during launch (this is discussed in more detail in Section 2 of Supplement 1).

Launch time t_0 was followed by vertical accelerations reaching 12.5 g. The Rb spectroscopy unit stayed locked during such accelerations, and the comb oscillator remained mode locked. After motor separation, just before the beginning of the microgravity phase, the two servo locks of the comb were turned on and a lock of f_{rep} was reached within seconds. For TEXUS 51, the microgravity regime ($<10^{-4} \text{ g}$ for all directions) started at $t_0 + 72 \text{ s}$, and the carrier offset frequency f_{ceo} was locked at $t_0 + 84 \text{ s}$. Such fast relocking of the comb was also achieved on the TEXUS 53 flight. The Rb beat note with the nearest comb line was counted until $t_0 + 330 \text{ s}$ [see Fig. 5(B)]. Then, for 120 s, a wavelength scan was initiated, resulting in the spectroscopy data shown in Fig. 4(B). To keep the duration of this scan short enough, a larger current step size was used compared to the precision scan in Fig. 4(A). From $t_0 + 450 \text{ s}$ on, the laser was locked again to the Rb line for 30 s. The microgravity phase ended at $t_0 + 442 \text{ s}$, directly followed by reentry into Earth's atmosphere. During re-entry, under decelerations of up to 12.5 g, the comb unit lost its phase locks, but the laser remained mode locked. The DFB laser module stayed locked to the Rb D₂ transition during both flights for the full flight period. Touchdown of the unpowered payload occurred at $t_0 + 884 \text{ s}$. The payload was recovered within a few hours and disassembled the same day. The comb module's functionality was positively assured within 24 h after return to ground.

The absolute clock comparison accuracy is limited by long-term drifts of the references, e.g., fluctuations of the locking point of the Rb laser, drifts of the CSAC, and some uncertainties due to light shift and Zeeman shift. Figure 5(A) shows a long-term dataset from ground. The systematic limits of our clock comparison are discussed in more detail in Section 4 of Supplement 1. However, for the relevant time scales below 1000 s, a relative frequency change can be measured with a precision approaching 10^{-11} at 100 s. Figure 5(B) shows the relative change in frequencies throughout the TEXUS 51 flight, including the data taken

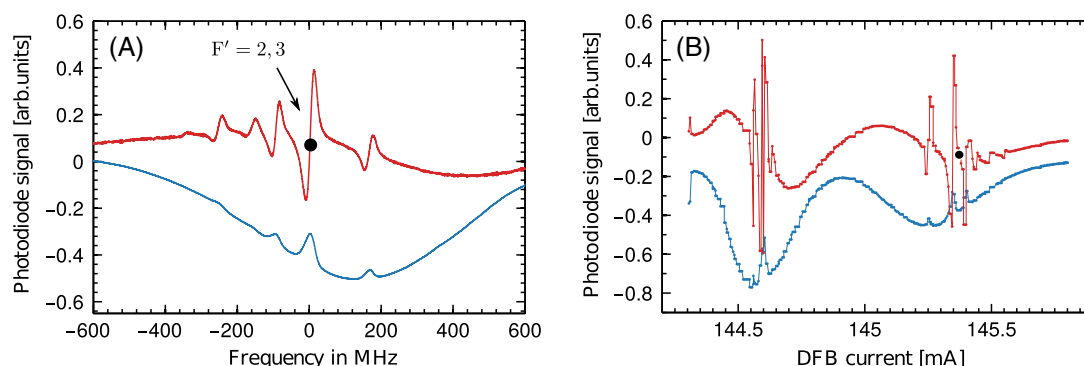


Fig. 4. (A) Plot of the spectroscopy signals obtained with the Rb module. The $^{87}\text{Rb } F = 2 \rightarrow F'$ transition manifold is shown. Blue, DC photodiode signal; red, demodulated signal. The plot shows data obtained on the ground, where the dot marks the locking point of the clock at 384.2279817(4) THz. (B) Comparable dataset from a current scan collected during the microgravity flight.

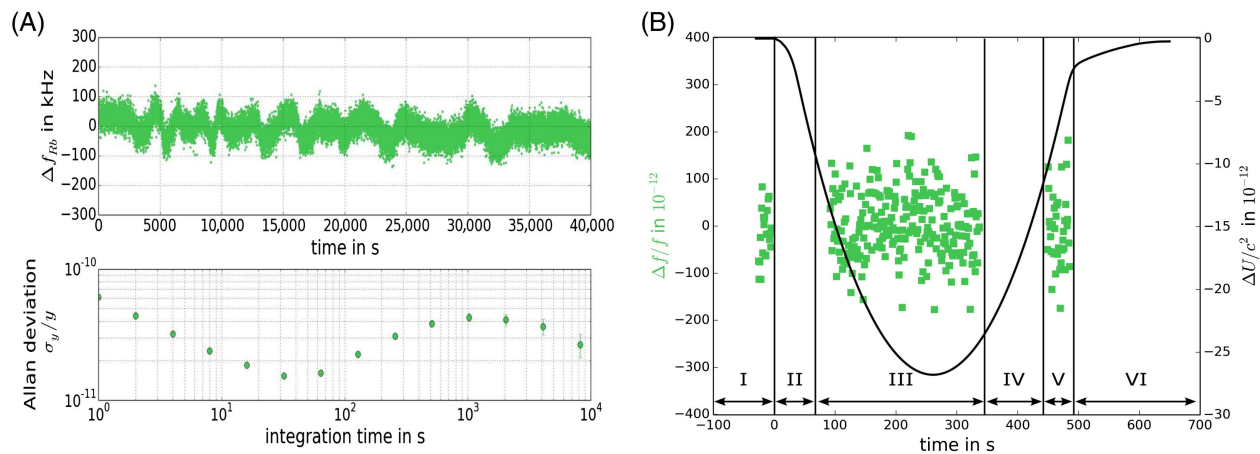


Fig. 5. (A) Time variation of the counted signal of the Rb spectroscopy over a period of 40,000 s on the ground, recorded at a gate time of 1 s. The resulting Allan deviation is plotted in the lower part of the figure. (B) Comparison of the frequency variation during the microgravity flight with respect to changes in the gravitational potential. The measurement sequence is indicated as follows: I, prelaunch data; II, launch; III, in microgravity locked to Rb D₂; IV, spectroscopy scan; V, in microgravity locked to Rb D₂; VI, re-entry.

immediately before launch, at payload apogee, and immediately before reentry into the atmosphere. A linear drift of 122 Hz/s, which was caused by a slight thermal drift within the spectroscopy module, has been subtracted from the flight data.

The clock comparison experiment can be evaluated with respect to the local position invariance (LPI), a cornerstone of general relativity, as it is one of the three postulates of Einstein's equivalence principle. If two different types of clocks are operating at different frequencies f_1 and f_2 , they can be compared in varying gravitational potentials ΔU , leading to $(f_1 - f_2)/f = (\beta_1 - \beta_2)\Delta U/c^2$, where β_i are dimensionless parameters depending on the sensitivity of each clock on LPI. Present clock comparisons on Earth [36] have determined that $\Delta\beta = \beta_1 - \beta_2 < 10^{-6}$. Taking into account recent clock accuracies reaching 10^{-18} [1,2], largely improved earthbound measurements are fundamentally limited by the gravitational variation given by the Earth's trajectory in the solar system. Performing a least squares analysis of the observed optical frequency during the sounding rocket flight and comparing it to the gravity potential dataset, as shown in Fig. 5(B), we evaluate $\Delta\beta < 0.19 \pm 0.26$. While we acknowledge that the present result is 5 orders of magnitude less precise than state of the art based determinations of $\Delta\beta$ on Earth, we note that neither the measurement principle nor the frequency comb is limiting us. For example, by choosing an advanced space-compatible RF reference like PHARAO [37,38] and an optical clock based on Yb, Sr, Al, or I₂ [39,40], a frequency comb based comparison on the level of 10^{-15} or beyond is feasible with a space comb on a satellite. We note that Menlo Systems has recently demonstrated an ultralow noise comb performance with an Allan deviation better than 10^{-16} at 1 s, dropping below 10^{-18} within 500 s [41]. Apart from additional low-voltage-driven electro-optical actuators in the oscillator and advanced high-bandwidth phase-locked loops, the fiber design of such a comb is very comparable and fits into a similar volume to that presented here. Such an instrument, on a trajectory passing through a gravitational potential difference $\Delta U = 10^{10} \text{ m}^2/\text{s}^2$ (i.e., from Earth to Mercury), could allow for the determination of $\Delta\beta$ down to a level of 10^{-9} , thereby

improving present Earth-bound experiments by 3 orders of magnitude.

4. CONCLUSION AND OUTLOOK

In conclusion, we have reproducibly demonstrated space-based determination of an optical atomic transition frequency, making use of a compact and remotely controllable comb setup referenced to a cesium clock. Our prototype experiment involved microgravity on a sounding rocket flight for a duration of 360 s. With respect to LPI, the apparent experimental limitations are caused by the reference clock, the selected Rb transition monitored using frequency modulation spectroscopy, and the limited variation in the gravitational potential during this flight. Nevertheless, with our prototype experiment, the technology readiness of optical frequency combs and remotely controlled precision laser spectroscopy has been significantly advanced. We stress that the current weight and volume of the module can be further optimized. Most of the current mass is caused by the structure itself. Moreover, in the future, the application of advanced atomic spectroscopy techniques like modulation transfer spectroscopy, combined with cavity enhancement, is feasible. We note that such devices have been recently designed for space use [40]. Such a spectroscopy setup could already improve the optical clock by 4 orders of magnitude to a stability of 10^{-15} . Finally, the electronics package can be improved while reducing the number of modules, power consumption, and performance and interface complexity. The next generation of the space comb setup with a significantly reduced weight below 10 kg, an overall power consumption below 30 W, and an improved spectroscopy module is currently under development and has been scheduled for a rocket launch in 2017. This next-generation system will also be tested for operation under vacuum conditions, and, moreover, some critical fiber components will be more deeply inspected for radiation hardness. Based on our technology, we expect a promising future for various comb applications in space. Space combs could largely serve for clocks and frequency standards, microwave generation, ranging, and laser calibration out there.

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See Supplement 1 for supporting content.

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